



# GoalFlow

## Features & Functionality

What the product does, and why each part exists

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**Stack** Flutter · Node.js + TypeScript + Express · MongoDB Atlas · FCM · Email

**Demo login** demo@goalflow.app / Demo1234

A personal goal-tracking mobile app. Define what you want to achieve, break it into milestones and actions, and see honestly whether you are on track.

**Flutter** (Android + iOS) · **Node.js + TypeScript + Express** · **MongoDB Atlas** · **FCM** (push) · **Email** (OTP, reset, digests)

|               |                              |
|---------------|------------------------------|
| Demo login    | demo@goalflow.app / Demo1234 |
| Screens       | 20                           |
| API endpoints | 53                           |
| Collections   | 7                            |

## 1. The product idea

Most goal apps are a to-do list with a due date. They tell you *what* you wrote down, never whether you are actually going to get there.

GoalFlow answers one question every time you open it: **"What should I focus on today, and am I on track?"** Everything in the product exists to answer that honestly — which means it will tell you when you are behind, and why, in a sentence you can act on.

The structure is three levels deep, because a goal you cannot start today is not a plan:

|           |                         |                          |
|-----------|-------------------------|--------------------------|
| Goal      | Learn Spanish           | 35% · On track           |
| Milestone | Build basic vocabulary  | 60%                      |
| Action    | Learn 20 new words      | 25 min · Mon–Fri · 07:30 |
| Action    | Complete one lesson     | 25 min · Mon–Fri · 07:30 |
| Milestone | Improve conversation    | 0%                       |
| Action    | Practise speaking aloud |                          |
| Action    | Review past vocabulary  |                          |

## 2. Onboarding

Five short steps, then one atomic commit. A user can never end up half-onboarded with a profile but no goal.

| Step | Collected                                                |
|------|----------------------------------------------------------|
| 1    | Name, optional profile photo                             |
| 2    | Main objective — the one thing this year is about        |
| 3    | First goal: title, why it matters, category, target date |
| 4    | Preferred days and time of day                           |
| 5    | Session length and weekly target                         |

Those answers are not stored and forgotten. They become:

- the default routine for every goal created later
- the default duration on every action
- the time reminders are scheduled for
- the tone of the home greeting — a morning person who opens the app at 4pm is told *"Your usual morning slot has passed — a short version still counts"*

Each new goal also arrives pre-filled with a **starter breakdown** for its category (Health, Learning, Career, Finance, Personal, Relationships, Productivity, Custom), so the user edits a plan rather than inventing one from a blank screen.

### 3. Goal management

Create, edit, **pause**, **resume**, **complete** and archive.

| Field                    | Notes                                              |
|--------------------------|----------------------------------------------------|
| Title, description       |                                                    |
| Why it matters           | Motivation is part of the product, not decoration  |
| Category                 | 8 built-in + custom                                |
| Priority                 | Low / Medium / High                                |
| Start and target date    |                                                    |
| Routine                  | Daily, specific days, N times per week, or one-off |
| Preferred time, duration | Inherited from onboarding, editable per goal       |
| Colour                   | Used across cards, calendar and charts             |

Two behaviours worth calling out:

- **Pausing stops future noise without touching history.** Upcoming actions are removed; everything already completed or missed stays exactly as it was.
  - **Deleting archives rather than destroys.** Progress you made does not disappear because you tidied up.
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## 4. The plan/log split

This is the decision the whole app rests on, and the reason it is not a CRUD app.

|                  | Role                      | Example                                    |
|------------------|---------------------------|--------------------------------------------|
| Action           | the rule                  | "Practise Spanish, 25 min, Mon–Fri, 07:30" |
| ActionOccurrence | one <b>dated instance</b> | "Wed 13 Aug, completed at 07:52"           |

A recurring action has no single status. "Gym on Mon/Wed/Fri" cannot answer "*did I miss Monday?*" if it is one row. Splitting the rule from the log makes it a lookup:

| Feature            | Becomes                                                                          |
|--------------------|----------------------------------------------------------------------------------|
| Today's actions    | one indexed query on today's date                                                |
| Missed detection   | a job flips <code>upcoming</code> → <code>missed</code> when the user's day ends |
| History            | occurrences are never deleted, only status-changed                               |
| Calendar           | occurrences already carry a date                                                 |
| Weekly progress    | count planned vs completed in the week                                           |
| Consistency streak | walk occurrence days backwards                                                   |
| Reminders          | every occurrence has a scheduled time — that <i>is</i> the queue                 |

Each occurrence carries one of five states: **upcoming**, **in progress**, **completed**, **missed**, **skipped**. A *skip* is treated as a decision, a *miss* as a failure — they are counted differently, because counting them the same punishes honesty.

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## 5. Home dashboard

One screen, one question. Served by a **single API call** so the app opens instantly rather than chaining five requests.

- Personalised greeting with what is left today
- Today's actions with one-tap completion
- **Yesterday's misses carried over for one day**, so catching up is easy

- 7-day consistency strip and current streak
- Active goals with progress ring and status chip
- Goals needing attention, with the reason spelled out
- Next milestone, recently completed items, unread notification count
- One-tap theme switch and notification bell in the header

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## 6. Progress tracking

Four levels, each shown where it is useful:

| Level     | Example                                           |
|-----------|---------------------------------------------------|
| Goal      | <i>Learn Spanish — 35% complete</i>               |
| Milestone | <i>Vocabulary — 18 / 30 actions completed</i>     |
| Weekly    | <i>This week — 22 / 33 planned actions</i>        |
| Long-term | 8-week trend, minutes invested, split by category |

Visualisations are used only where they add understanding: a **ring** for goal progress, a **7-dot strip** for the week, a **thin bar** for milestones. Charts are confined to the Progress screen so the rest of the app stays calm.

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## 7. Progress status — the part that is not a percentage

Every active goal is graded **Ahead** / **On track** / **Needs attention** / **Behind** / **Completed** from four independent signals:

```
timeRatio = how much of the goal's window has elapsed
workRatio = completed occurrences / all planned occurrences
adherence = last 14 days: completed / (completed + missed) [skips excluded]
delta      = workRatio - timeRatio
```

| Condition (first match wins)                                         | Status                                              |
|----------------------------------------------------------------------|-----------------------------------------------------|
| <code>workRatio &gt;= 1</code>                                       | Completed                                           |
| goal younger than 7 days                                             | On track — grace period, new goals are never shamed |
| <code>delta &gt;= +0.10</code>                                       | Ahead                                               |
| <code>delta &gt;= -0.05</code> and <code>adherence &gt;= 0.60</code> | On track                                            |
| <code>delta &gt;= -0.20</code> or <code>adherence &gt;= 0.40</code>  | Needs attention                                     |
| otherwise                                                            | Behind                                              |

Every status ships with a plain-language reason the UI shows verbatim:

*"You've completed 4 of 6 planned sessions this week." "Only 2 of your planned actions are done with 60 days left. Consider easing the routine rather than dropping the goal."*

A coloured chip on its own explains nothing. The sentence is the feature.

## 8. Consistency, without gamification

|                |                 |
|----------------|-----------------|
| Current streak | 20 days         |
| This week      | 22 / 33 planned |
| Last 30 days   | 71%             |

A day counts **only if something was planned for it**. Days with nothing planned are neutral — they neither break nor extend a streak. A rest day you scheduled yourself is not a failure.

There are no points, badges, levels or leaderboards. The number is there to be true, not to be maximised.

## 9. Calendar & schedule

A month grid showing planned, completed and missed actions plus upcoming milestones. Tap a day to see its actions, and complete, skip or **reschedule** from there.

It deliberately does not try to be Google Calendar. It exists only to make your goal-related schedule legible.

## 10. Weekly reflection

A dedicated screen, generated automatically each week:

- Goals worked on, actions completed and missed, completion rate
- Strongest area and the area needing attention
- Time invested, and next week's upcoming priorities
- Three optional prompts: **what went well** · **what was difficult** · **what to improve**

Reflections are saved per week and kept as history, so you can read back what you said three weeks ago and see whether you did anything about it.

## 11. Notifications

Everything is scheduled by the backend from the user's saved preferences. Nothing is hardcoded in the UI.

| Type                         | Channel      | When                              |
|------------------------------|--------------|-----------------------------------|
| Action reminder              | push         | N minutes before, N is user-set   |
| Daily summary                | push         | at the user's chosen morning time |
| Weekly digest                | push + email | user-chosen weekday and time      |
| Milestone reached            | push + email | on completion                     |
| <b>Login greeting</b>        | push         | on sign-in — see below            |
| Verify email, password reset | email        | transactional                     |

### The login greeting

The same "Welcome back" every time makes an app feel like it is not paying attention, so the message is chosen from how long you were away:

| Gap                | Message                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| First ever sign-in | <i>Welcome to GoalFlow, Priya</i> — set your first goal and we will turn it into something you can actually do each day |
| Same day           | <i>Hey again, Sam</i> — 4 things still open today                                                                       |
| 1–3 days           | <i>Welcome back, Sam</i> — 3 actions planned today, starting with Learn Spanish                                         |
| 4–14 days          | <i>Good to see you, Sam</i> — it has been 9 days, pick one and you are moving again                                     |
| 15+ days           | <i>Welcome back, Sam</i> — 40 days away, no catching up needed, just start from today                                   |

Copy for a long absence is written to remove guilt rather than add it. Shaming someone on the day they came back is how you lose them again.

## Preferences

Push and email toggles, reminder lead time, daily summary time, weekly digest day and time, milestone alerts, and a **quiet hours** window that correctly handles overnight ranges. Reminders and digests respect quiet hours; a greeting you triggered by signing in does not, because you are demonstrably awake.

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## 12. Appearance

**Light / Dark / System**, switchable from the home header in one tap or picked directly in Settings. The choice is stored on the device, survives sign-out, and applies instantly.

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## 13. Authentication

Sign up, email verification by 6-digit code, login, logout, password reset by emailed link, password change, and optional Google sign-in.

| Concern             | Approach                                                    |
|---------------------|-------------------------------------------------------------|
| Passwords           | bcrypt, cost 12, never returned or logged                   |
| Access token        | JWT, 15 minutes                                             |
| Refresh token       | Random, stored only as a SHA-256 hash, rotated on every use |
| Password change     | Signs out every device                                      |
| Account enumeration | Identical responses for known and unknown emails            |
| Brute force         | Rate limiting on every credential route                     |
| Authorization       | Ownership checked per resource                              |

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## 14. Personalisation summary

Everything the brief asks to be customisable is, and is used downstream:

preferred days · preferred time of day · exact start time · frequency (daily / specific days / N per week / one-off) · goal priority · action difficulty · estimated duration · weekly target · reminder lead time · progress style · quiet hours · personal constraints · theme

Two users with the same goal get genuinely different schedules, reminders and dashboards.

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## 15. Screens

| Group   | Screens                                                                      |
|---------|------------------------------------------------------------------------------|
| Auth    | Splash · Register · Verify email · Login · Forgot password                   |
| Setup   | Onboarding (5 steps)                                                         |
| Core    | Home · Today · Goals · Goal detail · Calendar · Progress · Weekly reflection |
| Editors | Create/edit goal · Create milestone · Create action                          |
| Account | Profile · Settings · Notification preferences · Notifications feed           |

## 16. Deployment

The backend and the database are hosted **separately**, on independent infrastructure:

| Piece      | Where                                                                            |
|------------|----------------------------------------------------------------------------------|
| API        | Its own dedicated server instance, deployed from source and running continuously |
| Database   | A separate managed MongoDB cluster, in its own region                            |
| Mobile app | Distributed as a release build                                                   |

Keeping them apart means the database is not tied to the lifecycle of the API server: the backend can be redeployed, restarted or scaled without touching the data, and the cluster is reachable only over an authenticated connection string held in server-side environment configuration — never bundled into the app.

## 17. End-to-end journey

Open app → Create account → Verify email → Onboarding

- Goal created with routine and starter breakdown
- Personalised dashboard
- Complete actions, watch progress and status update live
- Calendar and Progress views
- Weekly reflection
- Adjust the goal or routine, schedule rebuilds

Every step runs against the deployed backend with no placeholder screens.